

Industrial Cybersecurity

Defense in Depth & Cybersecurity Verification

Max Hsu

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Max Hsu

Lead SW Cybersecurity Testing Engineer

MaxMC.Hsu@moxa.com

2019

Joined Moxa

5+ years

Cybersecurity Experience

5+ years

Industry Experience

Experience

- ISA/IEC 62443 Expert
- Software Quality Assurance at Moxa
- Cybersecurity Testing Engineer at Moxa

Education

- National Cheng Kung University, Master of Telecommunication Management



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Agenda

1

Defense in Depth of OT

2

Defense in Depth of Equipment

3

Cybersecurity Testing

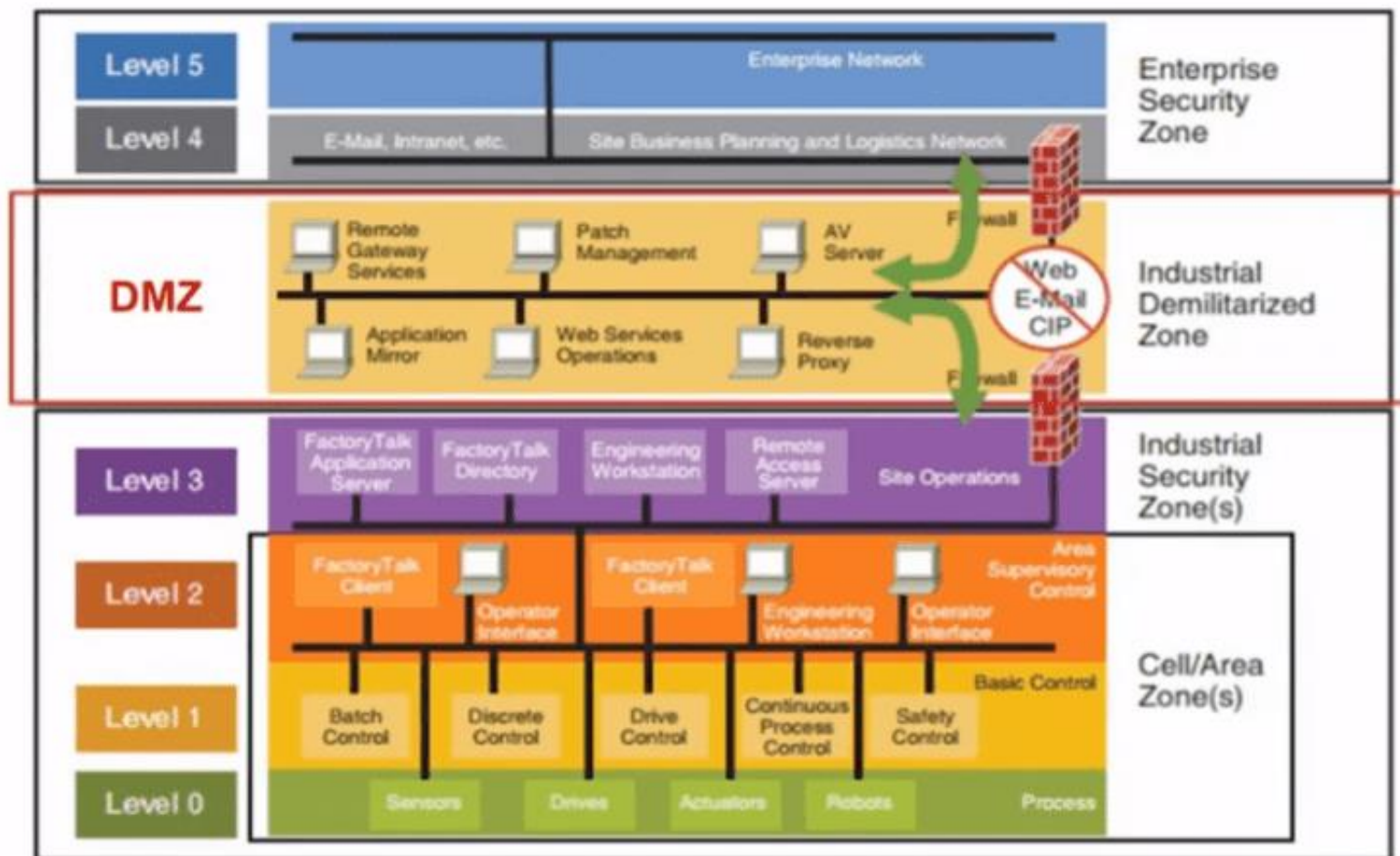
4

Case Sharing

The background of the slide features an aerial view of an offshore oil rig in the middle of a vast blue ocean. The rig consists of several red and white derrick structures. Overlaid on this image is a complex network of glowing blue lines and white circular nodes, representing a digital or industrial network. The right side of the slide is a solid teal color, which serves as a background for the title and logo.

Defense in Depth of OT

PERA (Purdue Enterprise Reference Architecture)



- 俗稱普渡模型 (Purdue Model)
- 1990 年代美國普渡大學 (Purdue) 提出的參考架構
- OT 層：Level 0 ~ 3
- IT 層：Level 4 ~ 5
- 部分認定沒有 Level 5
- 部分認定 DMZ 為 Level 3.5

圖片來源：packtpub.com

CIA Triad

- **Information Technology**

- **Confidentiality**
- **Integrity**
- **Availability**

- **Operational Technology**

- **Availability**
- **Integrity**
- **Confidentiality**



Defense in Depth of Operational Technology

- **Physical Protection**

- **Perimeter Security** : fences, gates, barriers, and motion sensors
- **Surveillance Systems** : CCTV, Security CAM
- **Security Personnel** : security staff
- **Access Control Systems** : ID cards, biometric scans (e.g., fingerprints or retina scans)



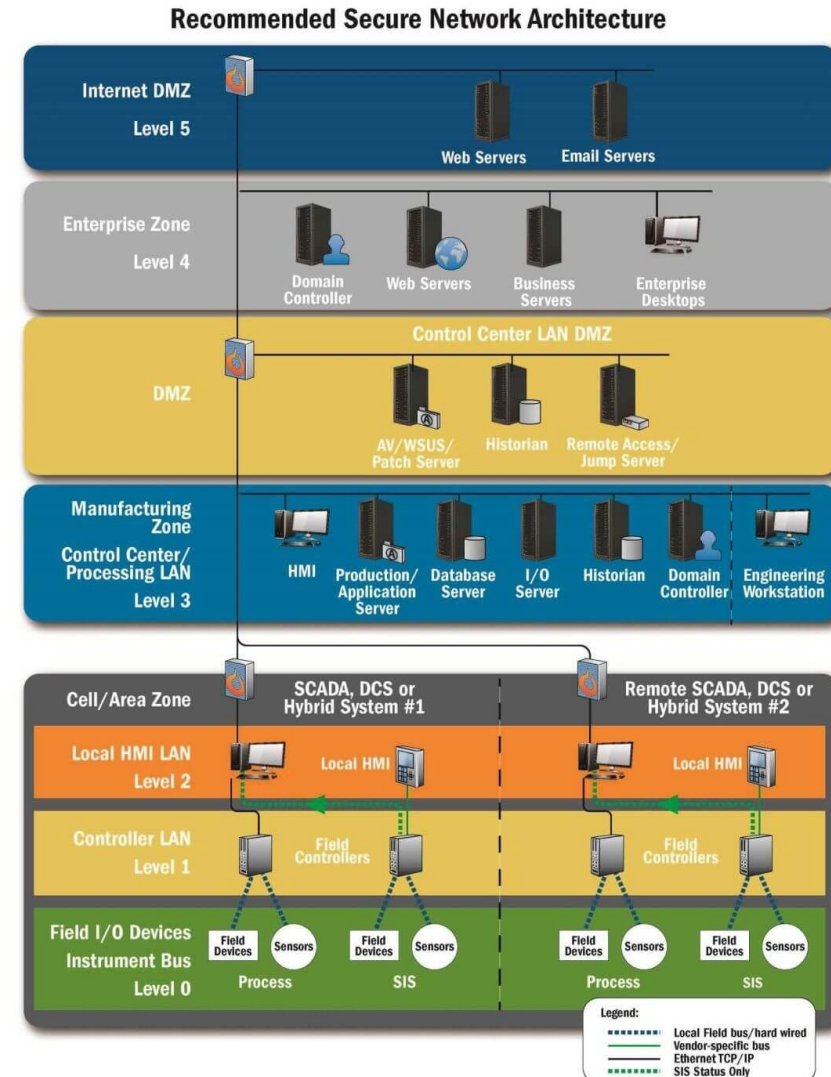
Defense in Depth for Operational technology

- **Logical Protection**

- Intrusion Detection Systems (IDS)
- Intrusion Protection Systems (IPS)
- Firewalls
- Antivirus and Antimalware
- Access Control Lists (ACLs)
- Network Segmentation
- Backup and Recovery Systems

[Is] it safe to install a firewall in front of every device?

Confidential



Department of Homeland Security (DHS) National Cybersecurity and Communications Integration Center (NCCIC) and Industrial Control Systems Cyber Emergency Response Team (ICS-CERT)
Recommended Practice: Improving Industrial Control System Cybersecurity with Defense-in-Depth Strategies

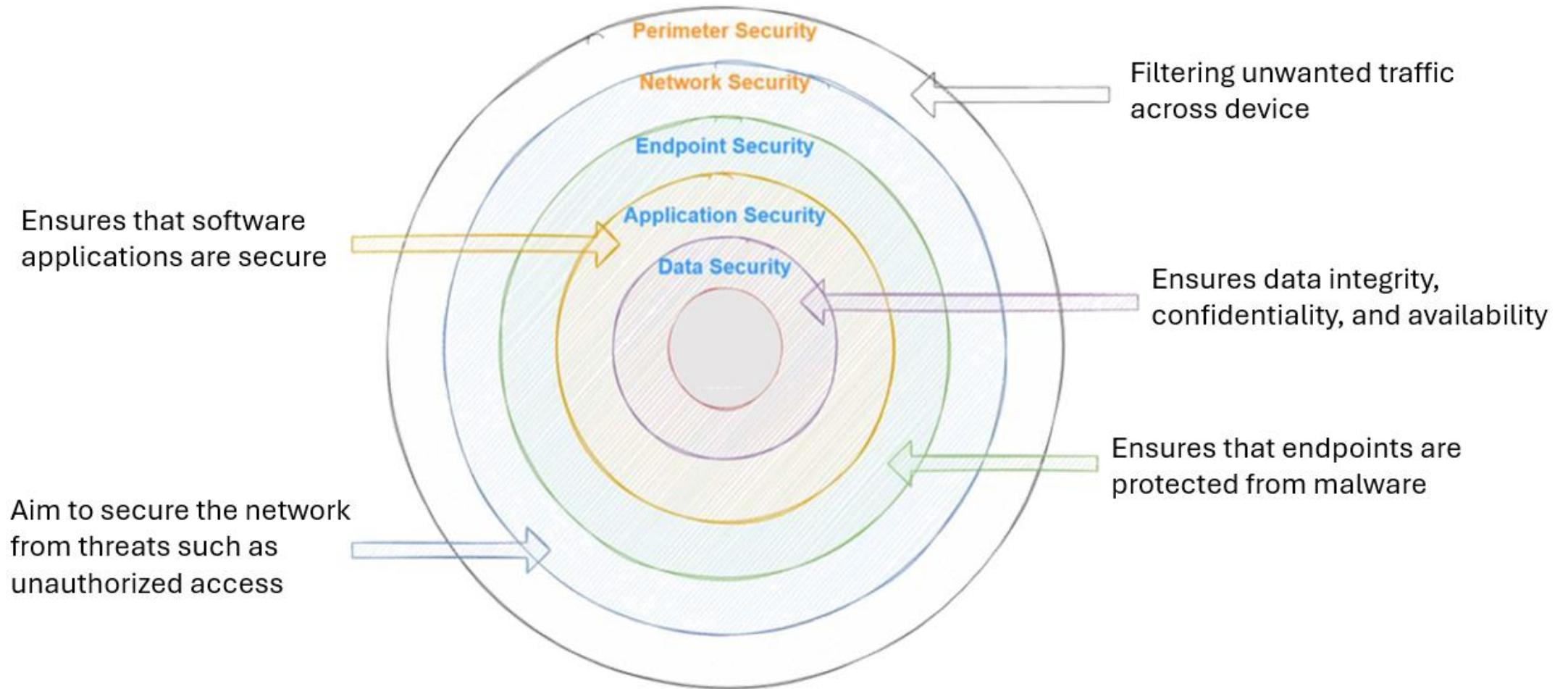
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Defense in Depth of Equipment



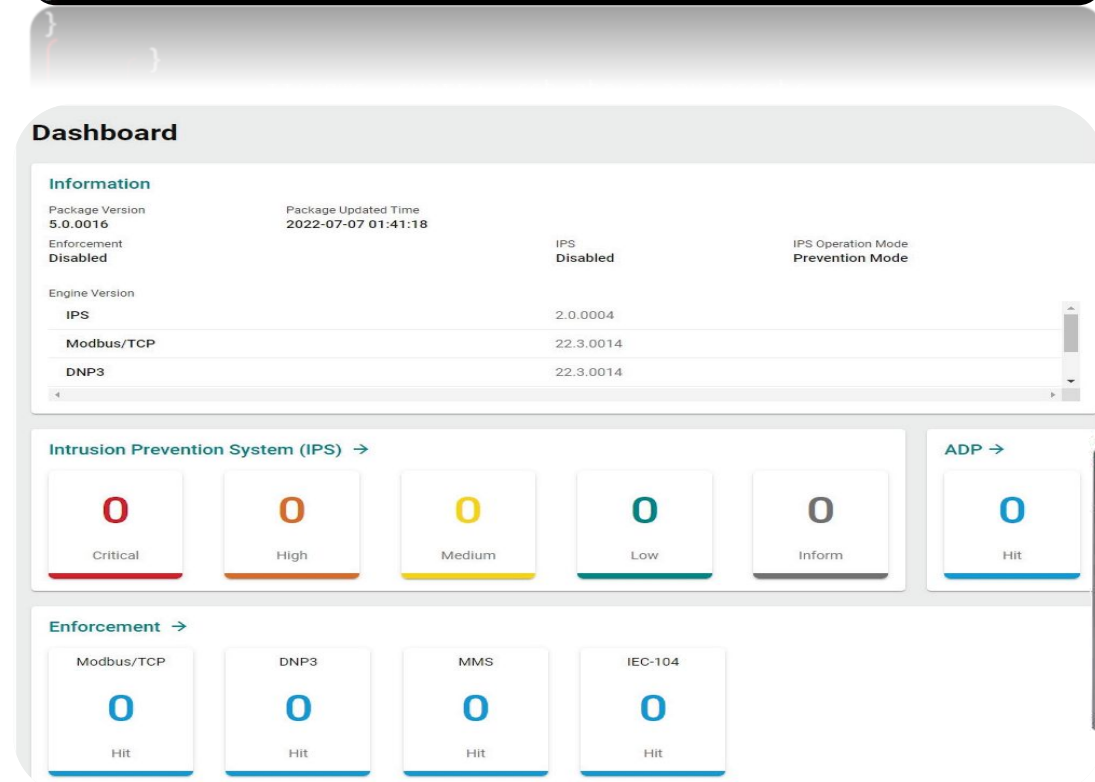
Defense in Depth of Equipment



Perimeter Security

- Physical Security
- Firewall
 - IPtables
 - Nftables
- Demilitarized Zone (DMZ)
- IPS/IDS
 - Signature Based
 - Anomaly Based

```
table inet table1 {  
    chain input {  
        type filter hook input priority 0; policy drop;  
        ct state established,related accept  
        iifname "ens192" udp dport openvpn counter packets accept  
        iifname "ens192" tcp dport ssh counter packets accept  
        iifname "ens224" tcp dport ssh accept  
    }  
}
```



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Network Security

- Network Segmentation

- Netmask
- VLAN
- VPN

- DoS Defense

- Null Scan
- Xmas Scan
- SYN/FIN Scan
- SYN/RST Scan

- Rate Limit

- SYN flooding
- ICMP flooding



NMAP

```
(root@kali)-[/home/kali]
# nmap -sN -vv 192.168.65.66
Starting Nmap 7.94SVN ( https://nmap.org ) at 2024-10-18 10:50 EDT
Initiating Ping Scan at 10:50
Scanning 192.168.65.66 [4 ports]
Completed Ping Scan at 10:50, 0.02s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 10:50
Completed Parallel DNS resolution of 1 host. at 10:50, 13.00s elapsed
Initiating NULL Scan at 10:50
Scanning 192.168.65.66 [1000 ports]
Completed NULL Scan at 10:50, 0.28s elapsed (1000 total ports)
Nmap scan report for 192.168.65.66
Host is up, received echo-reply ttl 254 (0.015s latency).
Scanned at 2024-10-18 10:50:39 EDT for 0s
All 1000 scanned ports on 192.168.65.66 are in ignored states.
Not shown: 1000 closed tcp ports (reset)

Read data files from: /usr/bin/./share/nmap
Nmap done: 1 IP address (1 host up) scanned in 13.41 seconds
Raw packets sent: 1004 (40.152KB) | Rcvd: 1001 (40.028KB)
```

```
Raw packets sent: 1004 (40.152KB) | Rcvd: 1001 (40.028KB)
hping3 done: 1 IP address (1 host up) scanned in 13.41 seconds
Raw packets sent: 1004 (40.152KB) | Rcvd: 1001 (40.028KB)
```

hping3
Network Scanner Tool

```
(root@kali)-[/home/kali]
# hping3 -S 192.168.65.200 --flood -a 192.100.100.100 -d 1024
HPING 192.168.65.200 (eth1 192.168.65.200): S set, 40 headers + 1024 data bytes
hping in flood mode, no replies will be shown
```

```
(root@kali)-[/home/kali]
# hping3 -I 192.168.65.66 --flood -d 1400
HPING 192.168.65.66 (eth1 192.168.65.66): icmp mode set, 28 headers + 1400 data bytes
hping in flood mode, no replies will be shown
```

Endpoint Security

- Secure Boot
- Encrypted Storage
- Rootfs backup/snapshot
- Secure Binary
- Software & File Integrity
 - AIDE
- Secure Upgrade
 - Integrity check
 - Authenticity check

(kali@kali)-[~]
\$ binwalk nport6000g2_release_no_boot_dev_V1.0.0_Build_24051615.bin

Binwalk

DECIMAL	HEXADECIMAL	DESCRIPTION
239024	0x3A5B0	DES SP2, little endian
239280	0x3A6B0	DES SP1, little endian
239664	0x3A830	SHA256 hash constants, little endian
257055	0x3EC1F	AES Inverse S-Box
257311	0x3ED1F	AES S-Box
2206708	0x21ABF4	Base64 standard index table
2207468	0x21AEEC	HTML document header
2208025	0x21B119	HTML document footer
2209072	0x21B530	XML document, version: "1.0"
2225643	0x21F5EB	Unix path: /etc/default/config/account-mgmt/product_account_mgmt.json for '%s'
2239548	0x222C3C	Base64 standard index table
2247204	0x224A24	Unix path: /etc/ssh/ssh_config
2251080	0x225948	Unix path: /etc/ssh/moduli
2376332	0x24428C	Unix path: /etc/ssh/ssh_known_hosts
2380539	0x2452FB	OpenSSH RSA1 private key, version "or opening %s: %s"
2381981	0x24589D	OpenSSH RSA1 private key, version "n OpenSSH private key (no footer)"
2466008	0x25A0D8	PEM certificate
2469784	0x25AF98	Neighborly text, "neighbors_idsystem"
2469800	0x25AFA8	Neighborly text, "neighbors_ports"
2469816	0x25AFB8	Neighborly text, "neighbors_port_descno"
2469836	0x25AFCC	Neighborly text, "neighbors_system"
2469856	0x25AFE0	Neighborly text, "neighbors"
2469953	0x25B041	Neighborly text, "neighborsR[%s][%d] "
2597120	0x27A100	SQLite 3.x database,
2633908	0x2830B4	SQLite 3.x database,



bash-5.1# ./checksec --dir=/bin											
RELRO	STACK	CANARY	NX	PIE	RPATH	RUNPATH	Symbols	FORTIFY	Fortified	Fortifiable	Filename
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	4	9	9	/usr/bin/gpg-wks-server
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	4	9	9	/usr/bin/mkfifo
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	4	9	9	/usr/bin/chcon
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	4	7	7	/usr/bin/pinky
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	5	17	17	/usr/bin/dir
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	3	6	6	/usr/bin/chmod
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	3	4	4	/usr/bin/logname
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	3	6	6	/usr/bin/base32
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	3	4	4	/usr/bin/pathchk
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	3	7	7	/usr/bin/chgrp
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	No	0	0	0	/usr/bin/gio-querymodules-64
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	5	13	13	/usr/bin/df
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	3	8	8	/usr/bin/readlink
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	5	11	11	/usr/bin/csplit
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	5	13	13	/usr/bin/grep
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	3	5	5	/usr/bin/sha224sum
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	3	5	5	/usr/bin/sha384sum
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	4	7	7	/usr/bin/factor
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	6	16	16	/usr/bin/cp
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	3	5	5	/usr/bin/sha1sum
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	3	5	5	/usr/bin/sha256sum
Full RELRO	Canary found	NX enabled	PIE enabled	No RPATH	No RUNPATH	No Symbols	Yes	6	9	9	/usr/bin/od

How to verify secure boot & disk Encryption

```
root@moxa:/home/moxa# systemctl status tpe.service
```

- tpe.service - MOXA ThingsPro Edge

Loaded: loaded (/lib/systemd/system/tpe.service; enabled; vendor preset: enabled)

Active: active (running) since Fri 2024-11-

01 17:42:07 CST; 20min ago

Main PID: 1358 (sysman)

Tasks: 49 (limit: 18983)

Memory: 256.3M

CPU: 23.205s

CGroup: /tpe.slice/tpe.service

└─1358 /usr/bin/sysman daemon

└─1519 /usr/bin/sysman audit daemon

└─1578 /usr/bin/sysman sms daemon

└─2302 /usr/bin/sysman app daemon

```
Nov 01 17:54:17 Moxa sysman[1358]: level=info msg="start tags monito
..."
```

```
root@moxa-1000059:/home/moxa# mx-interface-mgmt disk
```

NAME	DEVICE	SYSTEM_DISK	NUMBER_OF_PARTITIONS	AUTOMOUNT_SETTING	IO_STATE
SD	N/A	N	0	false	enabled
USB	/dev/sda	N	1	false	enabled
eMMC	/dev/mmcblk0	Y	4	false	N/A

```
root@moxa-1000059:/home/moxa# lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINT
loop0	7:0	0	262.8M	0	loop	/boot_device/p2/lower
sda	8:0	1	7.36		disk	
└─sda1	8:1	1	7.36		disk	

```
root@moxa-1000012:/home/moxa# lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINT
loop0	7:0	0	262.8M	0	loop	/boot_device/p2/lower
mtdblock0	31:0	0	1.3M	0	disk	
mtdblock1	31:1	0	2M	0	disk	
mtdblock2	31:2	0	2M	0	disk	
mtdblock3	31:3	0	64K	0	disk	
mtdblock4	31:4	0	64K	0	disk	
mtdblock5	31:5	0	1M	0	disk	
mtdblock6	31:6	0	4M	0	disk	
mtdblock7	31:7	0	4M	0	disk	
mmcblk0	179:0	0	14.76	0	disk	
└─mmcblk0p1	179:1	0	128M	0	part	/boot_device/p2/lower
└─mmcblk0p2	179:2	0	16	0	part	/boot_device/p2/lower
└─mmcblk0p3	179:3	0	12.56	0	part	/boot_device/p2/lower
└─mmcblk0p4	179:4	0	16	0	part	/var/log
└─p4_encrypted	253:1	0	1008M	0	disk	

Encrypted

Non-
Encrypted

Application Security

- **Authentication**

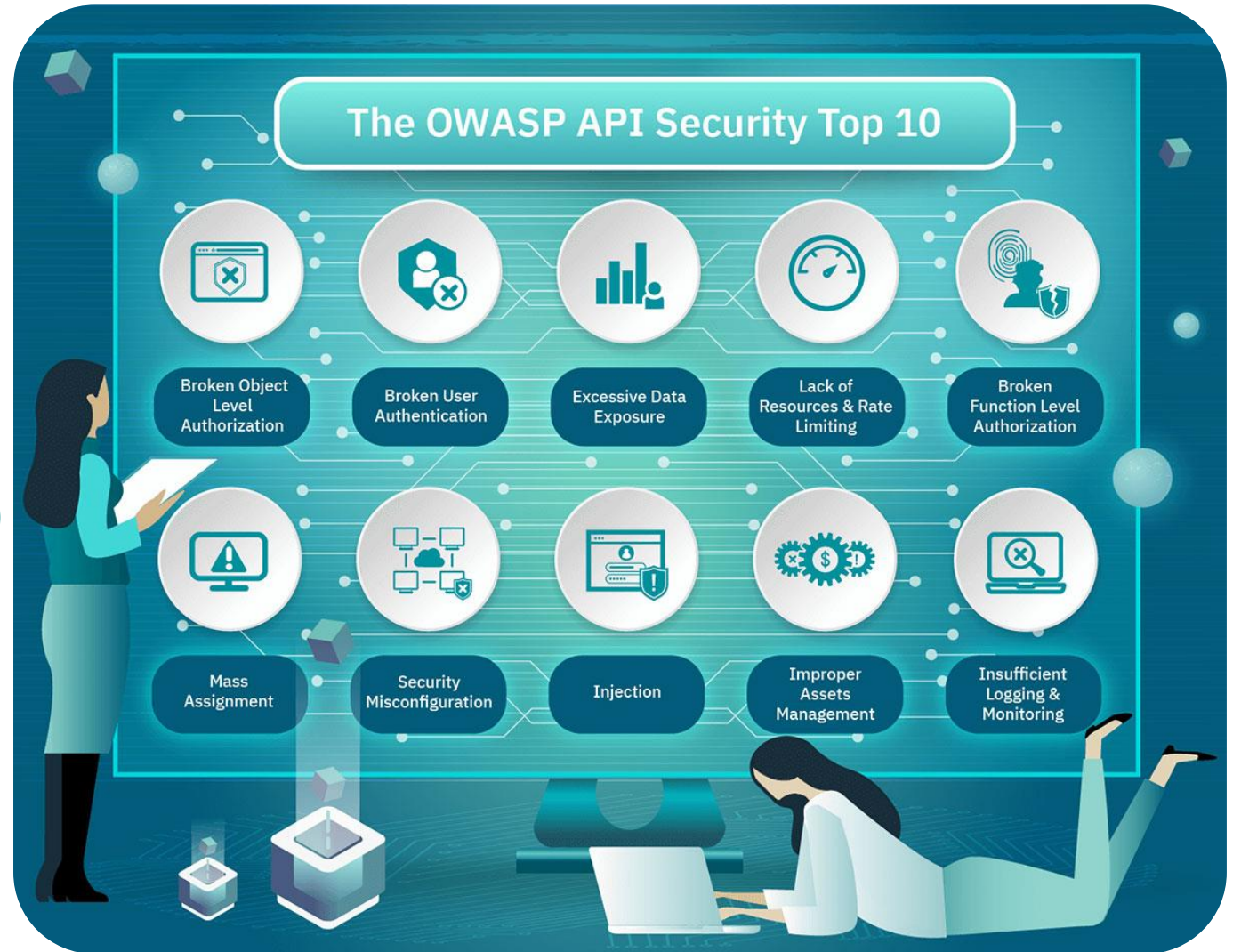
- Password
- Certification
- Radius/TACAS+ server

- **Authorization**

- Role-based Access Control (RBAC)
- Discretionary Access Control (DAC)
- Mandatory Access Control (MAC)

- **Audit Log**

- Timestamp
- Source
- Event



Data Security

- Data Encryption
- Secure Data Transmit Tunnel
 - TLS 1.2 above
 - SSH
- Hash File
 - md5sum
 - sha256sum
- Config Backup & Recovery

```
(root@kali)-[/home/kali/testssl.sh]
# ./testssl.sh -e 192.168.65.66

#####
testssl.sh      3.2rc3 from https://testssl.sh/dev/
(979a22c 2024-03-15 11:48:46)

This program is free software. Distribution and
modification under GPLv2 permitted.
USAGE w/o ANY WARRANTY. USE IT AT YOUR OWN RISK!

Please file bugs @ https://testssl.sh/bugs/

#####

Using "OpenSSL 1.0.2-bad (1.0.2k-dev)" [~183 ciphers]
on kali:./bin/openssl.Linux.x86_64
(built: "Sep  1 14:03:44 2022", platform: "linux-x86_64")

Start 2024-10-18 12:56:49      -->> 192.168.65.66:443 (192.168.65.66) <<--

rDNS (192.168.65.66):  --
Service detected:      HTTP

Testing 370 ciphers via OpenSSL plus sockets against the server, ordered by encryption strength

Hexcode  Cipher Suite Name (OpenSSL)      KeyExch.  Encryption  Bits   Cipher Suite Name (IANA/RFC)
-----
xc030    ECDHE-RSA-AES256-GCM-SHA384      ECDH 163  AESGCM      256     TLS_ECDHE_RSA_WITH_AES_256_GCM_SHA384
xc028    ECDHE-RSA-AES256-SHA384          ECDH 163  AES          256     TLS_ECDHE_RSA_WITH_AES_256_CBC_SHA384
x39      DHE-RSA-AES256-SHA               DH 1024    AES          256     TLS_DHE_RSA_WITH_AES_256_CBC_SHA
x35      AES256-SHA                       RSA       AES          256     TLS_RSA_WITH_AES_256_CBC_SHA
xc02f    ECDHE-RSA-AES128-GCM-SHA256      ECDH 163  AESGCM      128     TLS_ECDHE_RSA_WITH_AES_128_GCM_SHA256
xc027    ECDHE-RSA-AES128-SHA256          ECDH 163  AES          128     TLS_ECDHE_RSA_WITH_AES_128_CBC_SHA256
x33      DHE-RSA-AES128-SHA               DH 1024    AES          128     TLS_DHE_RSA_WITH_AES_128_CBC_SHA
x2f      AES128-SHA                       RSA       AES          128     TLS_RSA_WITH_AES_128_CBC_SHA
x16      EDH-RSA-DES-CBC3-SHA             DH 1024    3DES         168     TLS_DHE_RSA_WITH_3DES_EDE_CBC_SHA

Testing vulnerabilities

Heartbleed (CVE-2014-0160)      not vulnerable (OK), no heartbeat extension
CCS (CVE-2014-0224)             not vulnerable (OK)
Ticketbleed (CVE-2016-9244), experiment. not vulnerable (OK)
ROBOT                           not vulnerable (OK)
Secure Renegotiation (RFC 5746) supported (OK)
Secure Client-Initiated Renegotiation not vulnerable (OK)
CRIME, TLS (CVE-2012-4929)      not vulnerable (OK)
BREACH (CVE-2013-3587)         no gzip/deflate/compress/br HTTP compression (OK) - only supplied "/" tested
POODLE, SSL (CVE-2014-3566)     not vulnerable (OK)
TLS_FALLBACK_SCSV (RFC 7507)   No fallback possible (OK), no protocol below TLS 1.2 offered
SWEET32 (CVE-2016-2183, CVE-2016-6329) VULNERABLE, uses 64 bit block ciphers
FREAK (CVE-2015-0204)          not vulnerable (OK)
DROWN (CVE-2016-0800, CVE-2016-0703) not vulnerable on this host and port (OK)
```



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Cybersecurity Testing

Cybersecurity Testing

- Security testing is a form of non-functional software testing that checks the software for threats, risks, and vulnerabilities.
- Security testing focuses on a bunch of key elements such as,
 - Assets like applications, and infrastructure that need protection
 - Vulnerabilities that can expose the assets to attackers
 - Risk imposed by different security loopholes
 - Remediation of the security issues



Security Verification and Validation Testing

based on IEC 62443-4-1

- **Purpose**

- The **processes** specified by this practice are used to document the **cybersecurity testing** required to ensure that
 - All the **security requirements** have been met for the product.
 - Security of product is maintained when it is used in its product security context and configured to employ its **defense in depth strategy**.

Please prioritize these activities.

Security Requirement Testing

Vulnerability Scanning

Secure Code Review

Penetration Testing

Secure Code Review

Static / Dynamic Code Analysis



- **Identifying Vulnerabilities:**

- It helps in **detecting security flaws** such as SQL injection, cross-site scripting (XSS), and buffer overflows that could be exploited.

- **Preventing Costly Breaches:**

- By **catching vulnerabilities early**, secure code reviews can save time, money, and protect your reputation by preventing security breaches and data leaks.



Secure Code Review

Software Composition Analysis



- **Security:**

- SCA helps **identify vulnerabilities in open-source components**, which are commonly used in modern software development.

- **License Compliance:**

- Many open-source components come with specific **licensing requirements**. SCA tools ensure that your software complies with these licenses, avoiding legal issues.

- **Risk Management:**

- By creating a detailed inventory of all open-source components (often called a Software Bill of Materials or **SBOM**), SCA tools help manage and reduce risks associated with third-party dependencies.



Security Requirement Testing

Compliance Check



- **Identifying Security Gaps:**

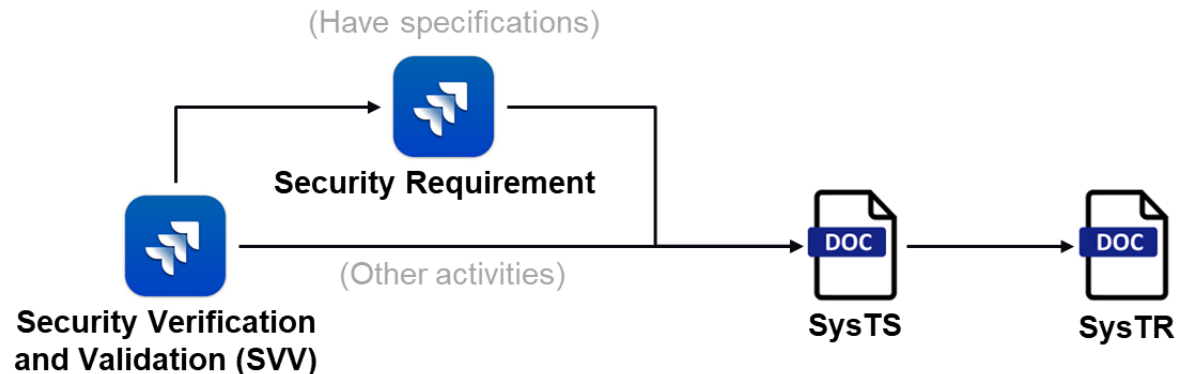
- It **ensures that all security requirements are correctly implemented** and that there are no gaps that could be exploited by attackers.

- **Compliance:**

- Many industries have **strict regulatory requirements** for security. Testing helps ensure that your software meets these standards, avoiding potential fines and legal issues.

- **Building Trust:**

- Demonstrating a commitment to security through rigorous testing helps **build trust with users and stakeholders**, showing that you take their data protection seriously.



Security Requirement Testing

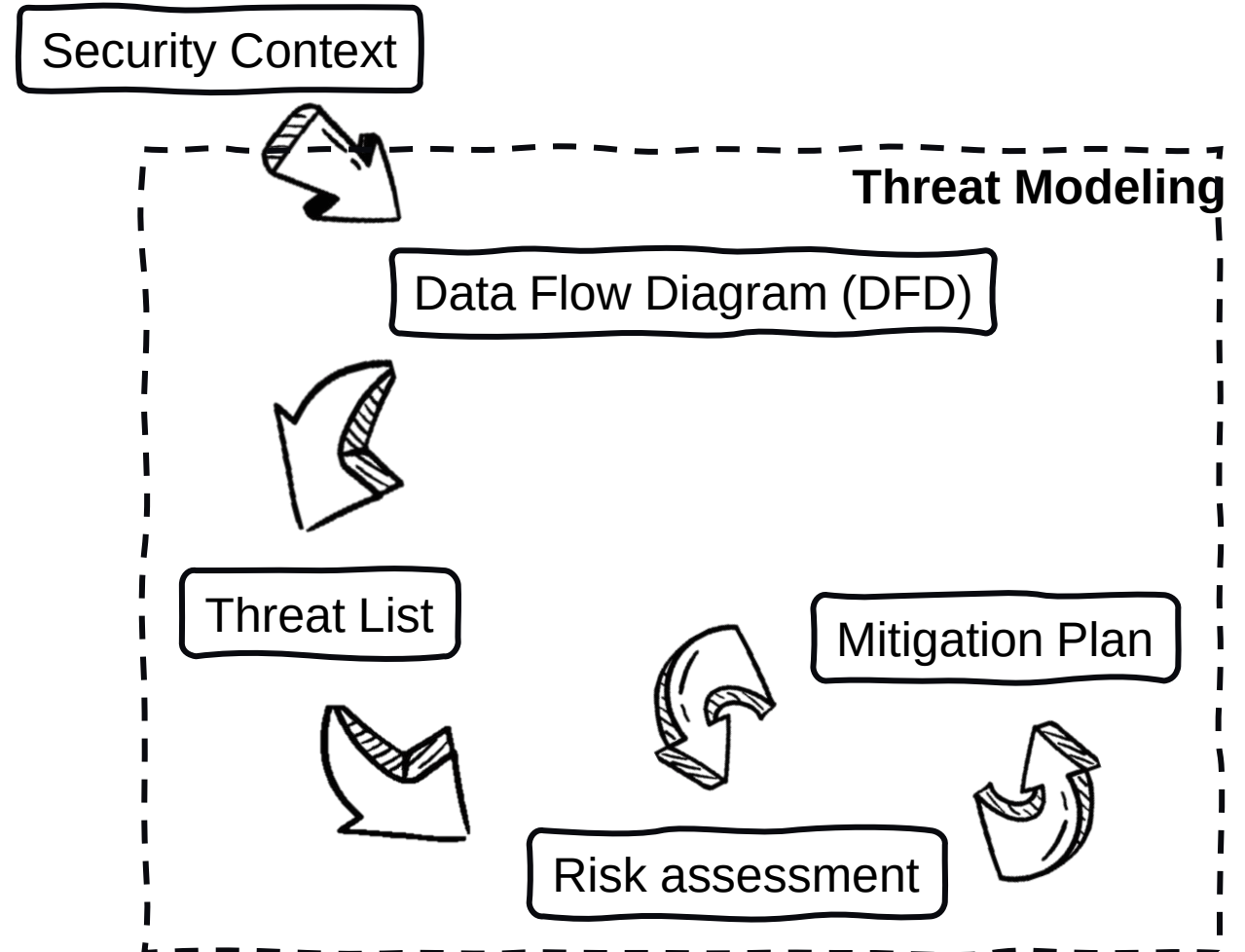
Threat Modeling

- **Risk Mitigation:**

- By identifying and addressing security vulnerabilities early in the development process, you can **significantly reduce the risk of security breaches** and their associated costs.

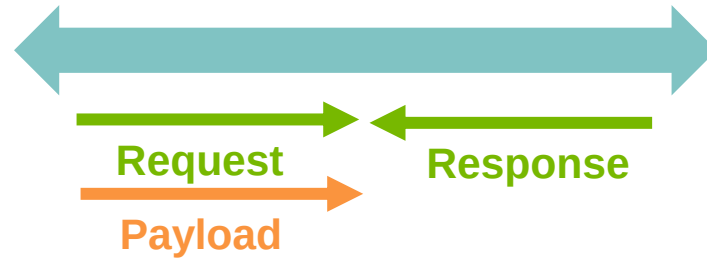


**Microsoft
Threat Modeling Tool**

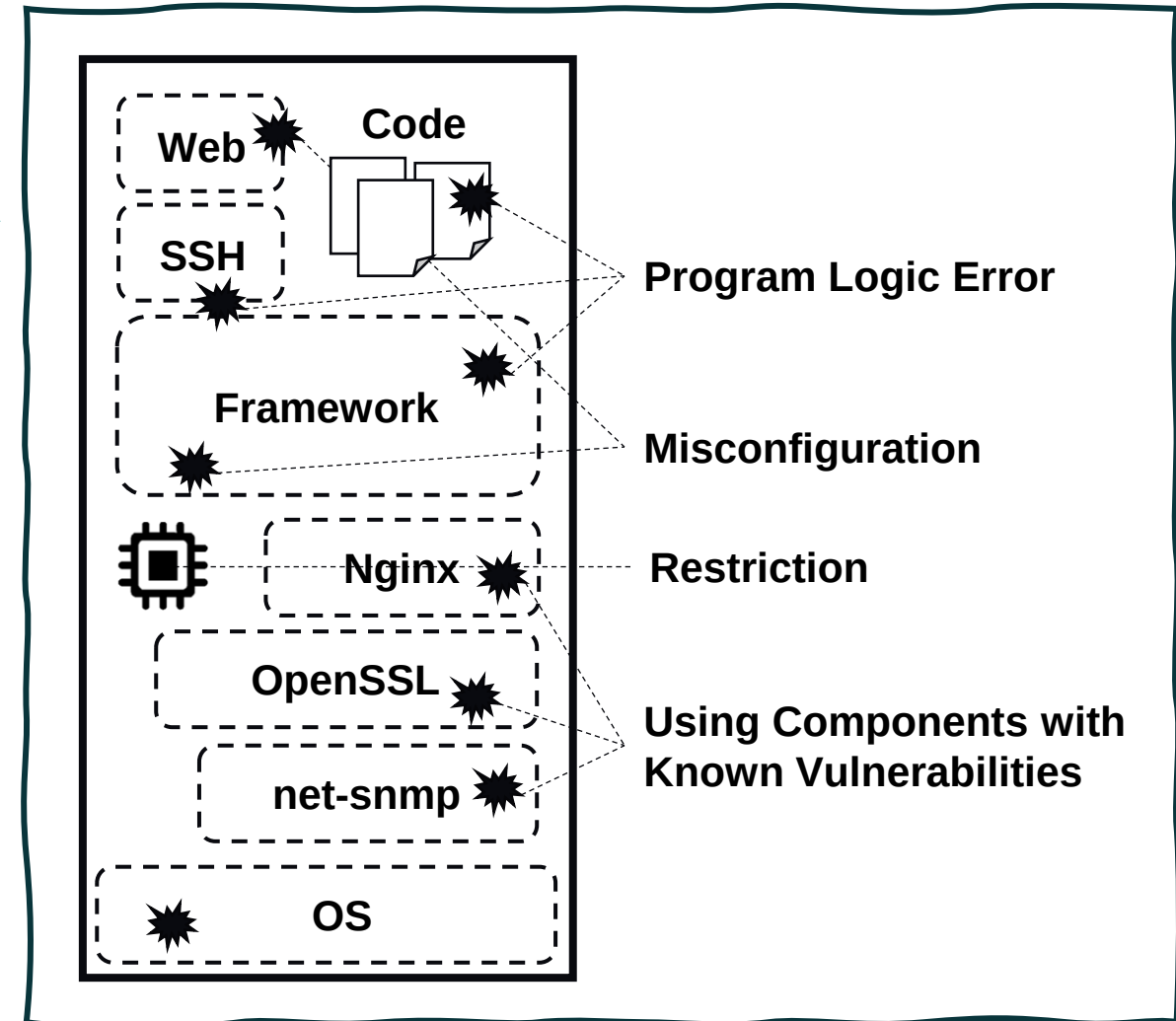


Vulnerability Testing

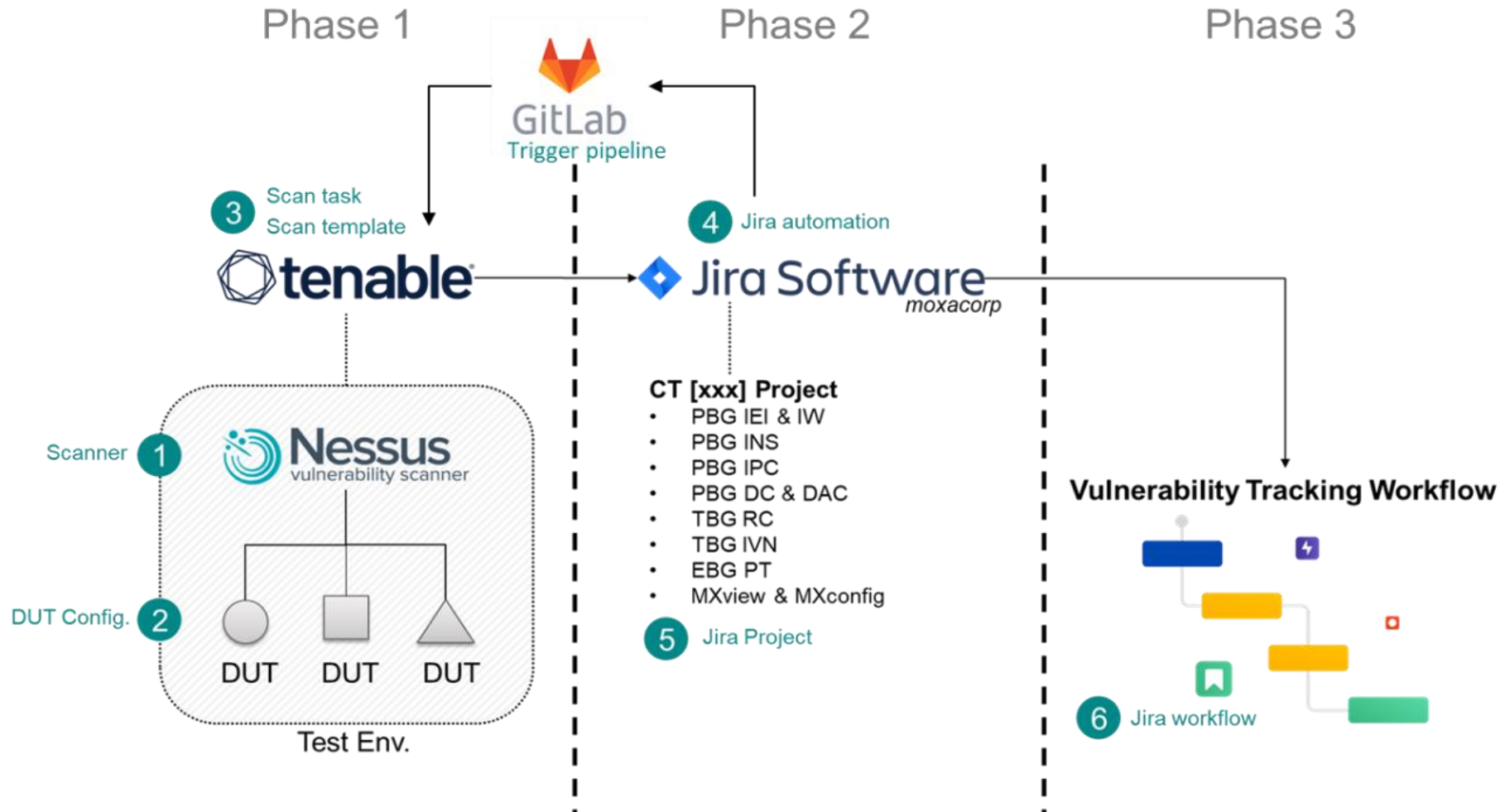
Black-box Vulnerability Scan



- (1) Let the target **produce unexpected abnormal behavior**
 - Crash, Reboot, Injection, ...etc.
- (2) Determine whether the vulnerability exists **based on** the information which **target responds**
 - Version, Header, ...etc.



DevOps in Nessus Scan



Vulnerability Testing

Fuzz Testing



- **Uncovering Hidden Bugs**

- Fuzz testing helps identify vulnerabilities and bugs that might not be found through traditional testing methods. By **inputting random or unexpected data** into a program, it can reveal edge cases and unexpected behaviors.

- **Enhancing Software Robustness**

- Fuzz testing ensures that software can handle unexpected inputs gracefully, improving its **overall robustness and reliability**.



Penetration Testing



- **Validating Security Measures**

- It verifies the effectiveness of existing security measures and controls. By **simulating real-world attacks**, pen testing ensures that security defenses are working as intended.



Metasploit

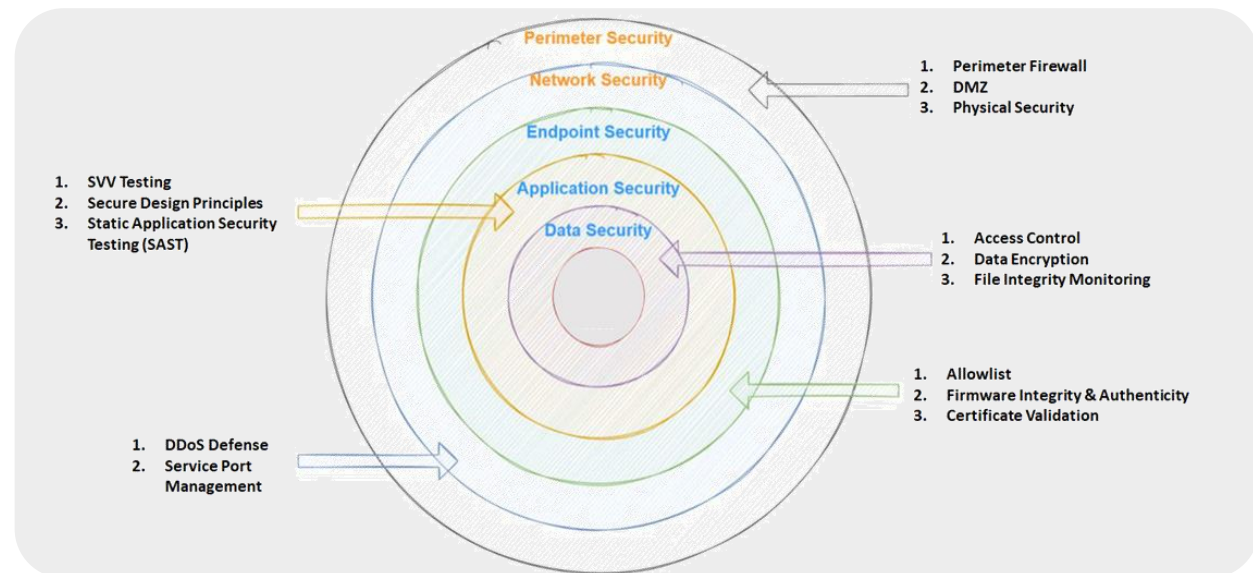


BURPSUITE

Summary

- 3 points to take a way

- CIA triad.
- Defense in Depth in cybersecurity.
- Security verification and validation in Industrial Product.



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Case Sharing

请勿模仿
DO NOT IMITATE

Case 1: 忘記linux的帳號密碼

1. Install john for crack password

```
(root@kali)-[/home/kali/testssl.sh]  
# apt install john
```

2. grep target user password in /etc/shadow

```
(root@kali)-[/home/kali/testssl.sh]  
# cat /etc/shadow | grep kali > shadow.txt
```

3. Crack by john

```
(root@kali)-[/home/kali/testssl.sh]  
# john --format=crypt shadow.txt  
Using default input encoding: UTF-8  
Loaded 1 password hash (crypt, generic crypt(3) [?/64])  
No password hashes left to crack (see FAQ)
```

4. Show cracked password

```
(root@kali)-[/home/kali/testssl.sh]  
# john --format=crypt shadow.txt --show  
kali:kali:19773:0:99999:7:::
```

反思：

1. /etc/shadow 夠安全嗎？
2. Password 複雜度的重要性
3. Password lifetime



Case 2 : 搶 Jay Chou 演唱會門票

1. 一張網卡連上WiFi AP，另一張網卡連上自己手機網路
2. 使用hping3 對WiFi AP 發送 flooding 攻擊



```
(root@kali)-[/home/kali]
# hping3 -S 192.168.65.200 --flood -a 192.100.100.100 -d 1024
HPING 192.168.65.200 (eth1 192.168.65.200): S set, 40 headers + 1024 data bytes
hping in flood mode, no replies will be shown
```

反思:

1. 不要相信任何人 (Zero Trust)
2. 控制自己可控制的
3. Network segmentation

3. 自己用手机網路那張網卡搶票



Moxa 企業參訪活動

Moxa 四零四科技邀請您

邀請各位前來 Moxa 總部，透過各種參訪活動與分享交流，一起一探 Moxa 的究竟！

- 參訪日期：2024/12/06(五) 9:30-16:00
- 地點：新莊宏匯廣場(新北大道四段3號13F)
- 請穿著方便行走的鞋子進行實驗室參觀活動
- 因應環保，請自備水壺
- 參訪當日備有午餐

報名截止日11/25(一)
人數有限，不要錯過機會囉！

報名連結



Moxa 四零四科技簡介

四零四是什麼樣的公司？有哪些重要的經營理念？
又是如何看待人才及培養人才？

產品及應用案例介紹

Moxa 的研發技術如何幫助客戶，提供有價值的服務？

實驗室參觀

追求品質的 Moxa 都是如何驗證我們的產品呢？
硬體與軟體測試實驗室

職務分享

軟、硬體研發高手雲集交流分享
他們都在公司做些什麼事呢？初生之犢如何站穩職涯腳步？

Moxa 生活

Moxa 都有些什麼活動，怎麼工作怎麼玩呢？

自由交流

其他的好奇或對未來職涯的迷惘，歡迎自由聊聊
履歷健檢、面試大補帖

Moxa社群

工作

生活

Moxa. work



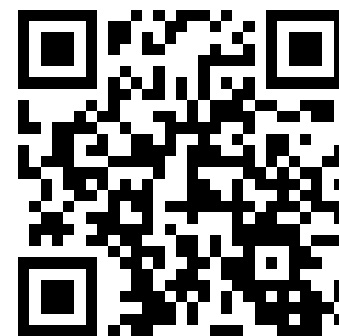
LinkedIn Moxa



moxa.career



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Thank You

For further information, visit www.moxa.com.

The Moxa logo is displayed in white on a teal background. It consists of the word "MOXA" in a bold, sans-serif font, followed by a stylized symbol that resembles a triangle or a stylized letter 'A'.

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